We describe a closed-loop framework for auditing and prioritizing improvement work across a corpus of engineering artifacts, and we document a sphere-aware extension of its curve-fitting stage together with the smallest audit unit that the framework admits. The framework has four composed techniques: (1) a learned latent curve, which re-expands a single 1-D ordering coordinate t into a fixed-width D-dimensional embedding via a Fourier basis with learned frequencies; (2) curve-guided recursive self-improvement, which uses sparse cells of the fitted curve as a prioritization lens for gap-mapping and bounded fixpoint-style editing cycles; (3) a hyperspherical-harmonic curve, which replaces the flat $[0,1]^2$ parameter manifold of (1) with the Riemann sphere S^2 and a learned Möbius $\phi_\theta \in \mathrm{PSL}(2,\mathbb{C})$ reparameterization; and (4) the single-action atom, which is the smallest unit of the loop — one corpus item maps to one point on S^2, one missing-primitive flip is one action, and the geodesic-only criterion gives an invariant that composes linearly across files. We give the governing equations for each technique, the empirical validation on a corpus of software skills ($N = 49$, $N = 70$, and $N = 79$ across three splits and one full-corpus audit), and a 474-dispatch atom experiment on the 79-skill corpus plus a 20-cycle experiment on 11 deep-research files, both showing zero negative Δ and confirming Lemma 1 and Theorem 1 directly.

Introduction

The paper is organized so each main-body section earns its place by improving fit, clarifying the geometry, or sharpening the claim. Sections 2–4 give the family background (flat curve model, hyperspherical variant, atom) at the level of governing equations. Sections 5–6 give the dataset, evaluation protocol, and headline results, with the 79-skill corpus audit reported as the primary empirical exhibit. Sections 7–8 are scope and conclusion. Two appendices carry the remaining material.

Background: The Learned-Latent-Curve Family

Flat curve model

For output dimension $j = 1, \dots, D$ (canonically $D = 384$) and 1-D coordinate $t \in [0, 1]$, the model is

$$z_j(t) = a_{j,0} + \sum_{m=1}^{k} \big(a_{j,m} \sin(2\pi f_m t) + b_{j,m} \cos(2\pi f_m t) \big),$$

with k shared learned frequencies $f_1, \dots, f_k$ and per-output coefficients $a_{j,m}, b_{j,m}$. Stacked over j, this is a curve $\gamma : \mathbb{R} \to \mathbb{R}^D$. Writing the design vector

$$\phi(t) = \big[\, 1,\; \sin(2\pi f_1 t),\, \cos(2\pi f_1 t),\, \dots,\, \sin(2\pi f_k t),\, \cos(2\pi f_k t) \,\big] \in \mathbb{R}^{1+2k},$$

the model is $\gamma(t) = C\, \phi(t)$ with coefficient matrix $C \in \mathbb{R}^{D \times (1+2k)}$, giving a parameter count of

$$P_{\mathrm{flat}} = D \cdot (1 + 2k).$$

At $D = 384$, $k = 8$: $P_{\mathrm{flat}} = 384 \cdot 17 = 6{,}528$ parameters.

Closed-form coefficient solve

With the frequencies f held fixed, the linear coefficient solve is the Tikhonov ridge,

$$C^{\star} = (\Phi^{\top}\Phi + \lambda I)^{-1} \Phi^{\top} Z,$$

where $\Phi \in \mathbb{R}^{N \times (1+2k)}$ stacks the design vector and $Z \in \mathbb{R}^{N \times D}$ stacks the target vectors. This is the sanity floor for any gradient-descent fit at the same frequencies: a fit worse than this at the same f means the optimizer failed, not the model.

Obtaining the coordinate t

The default pipeline sets t from the top principal component of a z-scored $N \times D_{\mathrm{feat}}$ feature matrix, mapped to [0,1]; a rank-uniformized variant replaces raw PC1 scores by their fractional ranks before mapping. The PC1+PC2 explained-variance ratio is the fit-quality gate (≥ 0.40); flat curves below the gate are not used.

The Hyperspherical-Harmonic Curve

Model

For $\mathbf{x} \in S^2 \subset \mathbb{R}^3$ (the Riemann sphere),

$$\begin{equation} \gamma(\mathbf{x}) = \mathbf{b} + \sum_{\ell=0}^{L} \sum_{m} \mathbf{a}_{\ell,m}\, Y^{S^2}_{\ell,m}\!\big(\phi_\theta(\mathbf{x})\big), \end{equation}$$

where $\mathbf{b} \in \mathbb{R}^D$ is the per-dim bias, $\mathbf{a}_{\ell,m} \in \mathbb{R}^D$ are the per-dim per-basis coefficients, and ϕ_θ is the Möbius reparameterization.

The parameter count is $D \cdot n_{\mathrm{basis}} + D + n_{\mathrm{M\ddot{o}bius}}$, where $n_{\mathrm{basis}} = \sum_{\ell=0}^{L}(2\ell+1)$ and $n_{\mathrm{M\ddot{o}bius}}$ is the real dimension of $\mathrm{PSL}(2,\mathbb{C})$ (Section [ref]). At $L=3$ on S^2, $n_{\mathrm{basis}}=16$; at $D=384$: $P_{\mathrm{sphere}} = 384 \cdot 16 + 384 + 6 = 6{,}534$.

Möbius reparameterization

$\phi_\theta(z) = (az+b)/(cz+d)$ with $a,b,c,d \in \mathbb{C}$ and $ad-bc=+1$. The four complex coefficients a,b,c,d carry 8 real components. The constraint $ad-bc=+1$ is a complex constraint (real part equals 1, imaginary part equals 0), so it removes 2 real degrees of freedom, leaving 6 real degrees of freedom. The $\mathrm{PSL}(2,\mathbb{C})$ identification (matrices M and $-M$ map to the same Möbius transformation) is a discrete identification and does not further reduce the real dimension. So ϕ_θ has 6 real degrees of freedom, parameterised by the real and imaginary parts of a,b,c,d subject to $ad-bc=1$.

We refine θ via L-BFGS-B with the closed-form ridge (Eq. [ref]) re-solved at each step. The basis is fixed; only the domain is reparameterized. Because every $\varphi \in \mathrm{PSL}(2,\mathbb{C})$ preserves the cross-ratio identically by definition, the cross-ratio check verifies implementation consistency: it fails if φ_θ is miscoded, not if the learned map is a poor fit.

The Single-Action Atom and Linear Composition

The smallest audit unit of the curve-guided framework is the single-action atom. We define it precisely so the only-positive-Δ invariant it carries propagates linearly across the corpus.

Atom: from file to S^2 point

For a single corpus item f (a file with $N_{\mathrm{sec}} \geq 2$ sections), the atom computes: \begin{enumerate} \item A per-section coverage vector $c_i \in \{0,1\}^9$, scored by pattern-matching against a fixed 9-D binary primitive basis $\mathcal{P} = (p_0, \ldots, p_8)$. \item A weighted aggregate $c = \sum_i w_i c_i$, with weights $w_i = \mathrm{len}(\mathrm{section}_i) / \mathrm{len}(f)$, thresholded at 0.5. \item A section coverage matrix M \in \{0,1\}^{N_{\mathrm{sec}} \times 9}, PCA top-2 via SVD giving (\bar u, \bar v) for the file aggregate. \item A stereographic lift \sigma: \mathbb{R}^2 \to S^2 with

\begin{equation} \sigma(u,v) = \frac{1}{1+u^2+v^2}\big(2u,\; 2v,\; u^2+v^2-1\big) \in S^2,

giving one point p = \sigma(\bar u, \bar v) \in S^2 per file. \item An ideal pole p^* = \sigma(\bar u^*, \bar v^*), where (\bar u^*, \bar v^*) is the lift of the all-ones coverage vector (1,\dots,1). \item A geodesic gap d(f) = \|p - p^*\|_2 (chordal proxy on S^2). \end{enumerate} For each missing primitive i \in \{j : c_j = 0\}, the atom simulates the flip c_i := 1, recomputes the S^2 point p', and selects i^* = \arg\min_i d_{\mathrm{post}}(f). This is the geodesic-only criterion: the chosen action strictly minimizes post-flip geodesic distance to the ideal pole.

Lemma 1 (atom invariant)

Lemma.

For any file f and any action \alpha selected by the geodesic-only criterion, \Delta_f = d_{\mathrm{pre}} - d_{\mathrm{post}} \geq 0.

Proof.

The criterion selects \alpha^* = \arg\min_i d_{\mathrm{post}} over the k candidates where k = |\{j : c_j = 0\}|. Each candidate flips exactly one missing primitive i from 0 to 1 and recomputes the file's S^2 point. If all k candidates had d_{\mathrm{post}} \geq d_{\mathrm{pre}}, the \arg\min would tie at d_{\mathrm{pre}} and \Delta_f = 0, never negative. A strictly negative \Delta_f would require d_{\mathrm{post}} > d_{\mathrm{pre}} for the \arg\min, contradicting the minimization. The action space is append-only (single-primitive flips); no candidate removes coverage.

Theorem 1 (linear composition)

Theorem.

For a corpus C with |C| = N files, every multi-file action \alpha_{\mathrm{corpus}} = (\alpha_1, \ldots, \alpha_N) where each \alpha_i is an atomic action on file f_i, has corpus-level

\begin{equation}

\Delta_{\mathrm{corpus}} = \sum_{i=1}^{N} \Delta_{f_i}.

\end{equation}

If every atomic \Delta \geq 0, then \Delta_{\mathrm{corpus}} \geq 0, and \Delta_{\mathrm{corpus}} > 0 if at least one atomic \Delta > 0.

Proof.

Each \alpha_i operates on its own file f_i independently: the per-file coverage matrix and S^2 point are unchanged for all f_j with j \neq i. The geodesic distance d_{f_i} is a function of f_i's coverage alone. Linear sum of non-negative scalars is non-negative.

M\"obius refinement strategy

The corpus-level \phi_\theta \in \mathrm{PSL}(2,\mathbb{C}) is one M\"obius transformation applied uniformly to all files; the atom-bound composition rule (Theorem 1) requires it stable across per-cycle atom dispatches, otherwise per-file \Deltas are measured in different charts and the linear sum does not apply to the reported cycles. Reported runs use the refine-once, \phi_\theta frozen mode

(ϕ_θ fit at corpus creation, frozen for all subsequent cycles); the joint-refine-per-cycle mode is reserved for $N_{\mathrm{items}} < 30$ or corpus growth $> 25\%$ since last refine.

Dataset and Evaluation Protocol

Dataset

The corpus consists of software-skill specifications (engineering artifacts) in the yubiOS repository. Each item has a 9-dimensional binary feature vector recording which of nine primitive capabilities the skill implements. We consider three splits:

- Split A (49 items): the first 49 items alphabetically. Train $N_{\mathrm{train}}=35$, holdout $N_{\mathrm{holdout}}=14$.
- Split B (70 items): all items at the time of the headline ablation. Train $N_{\mathrm{train}}=49$, holdout $N_{\mathrm{holdout}}=21$.
- Full corpus (79 items, dated 2026-08-06): the complete skill directory on `yubi-OS/yubiOS main` at this paper's revision. Used for the corpus-audit RSI in Section~6.3 and the 474-dispatch atom experiment in Section~6.4.

The 9-D binary feature space has principal-component concentration $\mathrm{PC1}+\mathrm{PC2} = 0.652$ at Split A and 0.548 at Split B; both clear the $\mathrm{PC1} \geq 0.40$ gate (Section [ref]).

Baseline

The capacity-matched baseline is a flat Fourier curve on $[0,1]^2$ with $k=2$ (the 2-D tensor-product extension of Eq. [ref]), giving $5 \times 5 = 25$ basis functions and $9{,}984$ parameters. We choose $k=2$ because it is the lowest-order 2-D tensor product that exceeds the 16-function count of the spherical variant (16 basis, $6{,}534$ parameters); $k=1$ gives only $3 \times 3 = 9$ basis and would not be a capacity match for the 16-function sphere, while $k=3$ gives $7 \times 7 = 49$ basis and ~ 24k parameters, which overwhelms the corpus. Both models are fitted with the closed-form ridge (Eq. [ref]) and the same ridge regularisation λ.

Evaluation metric

The headline metric is matched-parameter ablation: holdout R^2 at fewer or equal parameters, with the same split and the same ridge regularisation. We report the absolute holdout R^2 (not just the delta) so the result is interpretable against the corpus-mean baseline ($R^2 = 0$).

Reproducibility

All headline Split A and Split B numbers are single-run point estimates (no error bars) on a fixed holdout split, with a shared ridge regularisation λ across both arms. The audit-phase numbers (Sections 6.3) carry 5-seed $\pm$ std error bars for phases E--H, where the 5-seed multi-cycle stress test was run; the headline ablation does not. All audit runs reported in this paper use the 79-skill corpus dated 2026-08-06 (single dated snapshot, referenced throughout).

Results

Hyperspherical-harmonic variant: matched-parameter ablation on Splits A and B

The matched-parameter ablation on the two headline splits is the paper's single contribution: on these corpora, the hyperspherical parameter manifold is a strictly better inductive bias than the flat

[0,1]^2 baseline, with the absolute holdout R^2 positive on the smaller split and the relative \delta positive on both splits.

\textbf{Split A (49 items, N_{\mathrm{train}}=35, N_{\mathrm{holdout}}=14):}

• Hyperspherical model (S^2/L{=}3, 16 basis functions, 6,534 parameters): R^2_{\mathrm{holdout}} = +0.618.

• Flat Fourier baseline (k{=}2 on [0,1]^2, 25 basis functions, 9,984 parameters): R^2_{\mathrm{holdout}} = -0.359.

• Matched-parameter \delta: +0.977. The sphere wins by nearly 1.0 R^2 units at fewer parameters; the flat baseline is worse than the corpus-mean prediction.

\textbf{Split B (70 items, N_{\mathrm{train}}=49, N_{\mathrm{holdout}}=21, variant-included):}

• Hyperspherical model: R^2_{\mathrm{holdout}} = +0.222.

• Flat Fourier baseline: R^2_{\mathrm{holdout}} = -1.120.

• Matched-parameter \delta: +1.342. The sphere wins by more than 1.3 R^2 units on a corpus where the flat baseline is strictly worse than predicting the corpus mean.

Corpus audit across cycles 5--9 (phases A--H)

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\textbf{Figure 1.

\centering \includegraphics[width=5.5in]{chart-A-H-table-1-headline.png} \caption{Table 1. Headline numbers across cycles 5--9 (phases A--H; 70\rightarrow73-skill corpus).}

Atom experiment: 474 dispatches on the 79-skill corpus + 20 cycles on the 11-file deep-research corpus

The 79-skill corpus was audited via 474 atom dispatches across 6 cycles (single run on the 2026-08-06 snapshot, commit \texttt{6ae3abeb65} on `yubi-OS/yubiOS main`). Cumulative \Delta = +11.2971 across all dispatches; zero negative \Delta; 116 strictly positive \Delta. Sparse cells: 7 \to 3 \to 2 \to 3 \to 0 \to 0. Fixpoint reached at cycle 6 (peak \Delta below the 0.001 epsilon). The corpus reached 79/79 = 100\% coverage across all nine primitives.

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\textbf{Figure 1.

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\textbf{Figure 1.

The 11-file deep-research corpus was audited via 20 cycles (initial sweep + 7 post-edit re-fits). Across both corpora combined, the geodesic-only criterion produced zero negative \Delta --- 494 dispatches in total, 0 regressions --- confirming Lemma 1 directly. Peak \Delta trajectory on the 11-file corpus: +0.3092 (cycle 2, advisor-report) \to +0.2810 (cycle 8, pkcs11-ecdsa-deepdive) \to +0.1872 (cycle 14, falco). Peak \Delta reduced 39.5\% across three peak runs, mean \Delta reduced 28.2\%, cumulative \Delta plateaued at +1.6882. Per-file \Delta reductions: advisor -55.7\%, pkcs11-ecdsa-deepdive -66.6\%, comparative -100\% (converged to local minimum). Two of the eleven deep-research files (comparative-V52-refresh and debugging-journal) reached local minimum coverage shape (\Delta = 0) and required no further action. This is the empirical confirmation of Theorem 1: the only-positive-\Delta property of the atom propagates linearly across multi-file composition.

Discussion

What this result does and does not show

The hyperspherical-harmonic variant wins on the matched-parameter ablation at both Splits A and B by a margin (Split A \delta = +0.977, Split B \delta = +1.342) that is hard to attribute to noise. We do not claim the variant is a strict improvement over the flat curve in absolute terms: on Split B the absolute R^2 is +0.222 (positive but small), and the headline numbers are single-run point estimates without error bars. The atom experiment (Figure 2 + Figure 3) shows the smallest audit unit composes without regression --- 494 dispatches, 0 negative \Delta --- but does not by itself validate the variant.

Limitations

The empirical case is fragile because it is one corpus family, one target, and the headline numbers are single-run point estimates. The 5-seed error bars in Figure 1 (phases E--H) partially address the empirical-fragility concern for the audit; the headline Split A/B numbers do not. The 79-skill corpus audit (Section 6.4) uses 474 dispatches across 6 cycles at N=79 --- enough to support an empirical claim at the atom-invariant level, but not enough to support multi-seed error bars at the headline-ablation level. A synthetic-manifold benchmark (one not run) and a second-corpus re-run would replace the present single-seed point estimates with error bars at both levels.

Conclusion

The hyperspherical-harmonic curve replaces the flat [0,1]^2 parameter manifold with the Riemann sphere S^2 and learns a M\"obius reparameterization \phi_\theta \in \mathrm{PSL}(2,\mathbb{C}) of the domain. On the yubiOS software-skill corpus, it wins on the matched-parameter ablation at both Splits A and B (Splits A \delta = +0.977, Split B \delta = +1.342), with fewer parameters and no error bars. The single-action atom is the smallest unit of the resulting audit pipeline; its only-positive-\Delta invariant propagates linearly to multi-file composition (Theorem 1), as confirmed by a 474-dispatch experiment on the 79-skill corpus and a 20-cycle experiment on 11 deep-research files showing zero negative \Delta across the combined 494 dispatches. Future work: a synthetic-manifold benchmark would test whether the inductive-bias claim holds on manifolds other than S^2, and a second-corpus re-run would replace the present single-seed point estimates with error bars.

Appendix A: Atom Coverage of 79 Skills (Empirical)

The 79-skill corpus reached 100\% primitive coverage (79/79 for all nine primitives) after 6 RSI cycles. Per-primitive coverage progression: cycle 1: [71, 63, 63, 49, 35, 74, 72, 70, 73]; cycle 6: [79, 79, 79, 79, 79, 79, 79, 79, 79]. Cumulative \Delta across 474 dispatches: +11.2971, with 116 strictly positive single-action flips and zero negative \Delta --- the empirical verification of Lemma 1's only-positive-\Delta invariant on a corpus 7\times larger than the 11-file audit. Sparse cells: 7 \to 3 \to 2 \to 3 \to 0 \to 0. Fixpoint reached at cycle 6.

Appendix B: Cycles 10--15 RSI Run on Full Repo (Future Work)

The 79-skill audit in Section 6.4 used cycles 10--15 of the curve-guided-rsi self pipeline (cycles 1--9 were the prior phases A--H audit on the 70-skill corpus). The full multi-corpus audit, including differential curve baselines across the self-doc corpus and the engineering corpus, is future work.

References

{durastanti2026} Durastanti, C. (2026). Spectral Bayesian Regression on the Sphere. arXiv:2601.20528 [math.ST]. https://arxiv.org/abs/2601.20528

{anonymous2022} Anonymous (ICLR 2022 under review). Generalized Fourier Features for Coordinate-Based Learning on Manifolds. OpenReview \texttt{g6UqpVislvH}.

{rahimi2007} Rahimi, A., \& Recht, B. (2007). Random Features for Large-Scale Kernel Machines. NeurIPS 2007.

{tancik2020} Tancik, M., et al. (2020). Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains. NeurIPS 2020.

{sitzmann2020} Sitzmann, V., Martel, J., Bergman, A., Lindell, D., \& Wetzstein, G. (2020). Implicit Neural Representations with Periodic Activation Functions. NeurIPS 2020 (SIREN).

{mildenhall2020} Mildenhall, B., et al. (2020). NeRF: Representing Scenes as Neural Fields for View Synthesis. ECCV 2020.

{cohen2018} Cohen, T. S., et al. (2018). Spherical CNNs. ICLR 2018.

{nickel2017} Nickel, M. \& Kiela, D. (2017). Poincar\'e Embeddings for Learning Hierarchical Representations. NeurIPS 2017.

{ahlfors1979} Ahlfors, L. V. (1979). Complex Analysis, 3rd~ed. McGraw-Hill.

{docarmo1976} do Carmo, M. P. (1976). Differential Geometry of Curves and Surfaces. Prentice-Hall.

{smith2026} Smith, E. (2026). Learning in Curved Weight Space: Exponential-Linear Weight Reparameterization for Improved Optimization. arXiv:2607.09967 [cs.LG].